production. If sufficent time is given, any operator would work and maintain the sizes with in a close degree of accuracy. Hence, tolerance were introduced, It helps to increase the production and to achieve the required fits. The same standards follow all over world which helps their sales internationally.

For Examples:

Bolts, Nuts, Threads, Keys, Studs, Washers, Tapers, Gears, Bearings and different sizes of plates and wires are available in standard sizes very easily. The production is also increased by reducing the cost. Then only the components are easily available sole everywhere. Trading from one country to another country is also increased.

8.2 STANDARDIZATION

Standardization is the process of implementing and developing technical standards based on the consensus of different parties that include firms, users, interest groups, standard organisations and governments. It helps to maximize compatibility, interperability, safety, repeatability and quality.

If any component is broken or to be changed due to wear and tear, it can be easily replaced by a new components by purchasing it from market. These parts are manufactured by following the quality, the accuracy of size and other standards strictly prescribed by the following organisations like ISO and BIS. This is called Standardization.

8.3 INTERCHANGEABILITY



Interchangeability parts, the ability to select components for assembly at random and fit them together within proper tolerances.The ability that an object can be replaced by another object without affecting code using the object.

In making mating parts in mass production, it is not always possible to maintain the dimensional accuracy of the parts. There are some variations in dimensions in few of the many number of products. Variations of sizes are within certain limits. Certain deviations are allowed to ensure Interchangeability of mating parts.

8.4 ADVANTAGES OF INTERCHANGEABILITY

1. The assembly of mating parts is easier.
2. The rate of waste in mass production becomes less.
3. It reduces assembly time and cheaper by employing unskilled worker.
4. Random assembly of components is possible.It reduces cost of production.

8.5 BASIC TERMINOLOGY IN INTERCHANGEBLE SYSTEM

Shaft: The shaft indicates the outer diameter of a cylinderical profile, but also

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